

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

- a. Participant Name: Lin Jie-Rou (Astrid)
- b. Problem Statement: AI for Better Living and Smarter Communities — create a data intelligence tool people would actually use, and show how acceleration helps them make a faster or better decision.

Brief about the idea

An elderly parent's health data is scattered across three disconnected places: clinic prescriptions, home blood-pressure/glucose readings, and diet or symptom notes the family shares in a group chat. No one connects the dots until something goes wrong.

CareSignal pulls all three sources into BigQuery and runs them through a clinical rule engine and risk-scoring model built on real drug-nutrient-vital sign interaction logic — from the founder's own clinical nutrition background, not a general-purpose LLM guess. When risk crosses a threshold, Gemini turns the finding into a plain-language alert within seconds, with a full audit trail back to the exact data point and rule that triggered it.

Built for family caregivers coordinating care from a distance — not another hospital dashboard.

Your solution should be able to explain the following:

- How did you approach the common problem statement and translate it into a working solution using Google Cloud technologies (e.g., BigQuery, ADK, Vertex AI, Gemini, etc.) while ensuring the hackathon's core requirements are met? If you selected the NVIDIA problem statement, also explain how NVIDIA technologies were integrated into your solution.
 - What real-world problem does your solution address, and what practical impact does it create for users, organizations, or communities??
 - What is the core architecture or workflow behind your solution, and how does it transform data into actionable insights or intelligent decisions?
- **Approach:** Pub/Sub + Cloud Functions land multi-source caregiver data in BigQuery. A clinical rule engine plus a BigQuery ML / Vertex AI risk scoring model evaluates it. Vertex AI Pipeline orchestrates the ingest-score-explain flow; Gemini turns a risk finding into a plain-language, source-cited alert — core requirements (BigQuery, Vertex AI, Gemini) all met.
- **Real-world impact:** Family caregivers manually cross-check scattered medication, vitals and diet data today — 15-30 minutes, and risks are often missed by non-clinicians. CareSignal cuts that to under 5 seconds, for the home-caregiver segment most long-term-care AI tools ignore.
- **Workflow:** Raw events flow into BigQuery, get scored by the rule + ML engine, and any flagged risk is explained by Gemini with a drill-down audit trail back to the exact rule and data point — turning disconnected numbers into one actionable decision.

Opportunities

- How different is it from any of the other existing ideas?
- USP of the proposed solution

Most long-term-care AI targets institutions (hospitals, nursing homes) and hands the actual judgment to a general LLM prompt. CareSignal targets the family caregiver — an underserved, community-level user — and keeps the judgment in an explainable clinical rule and risk-scoring engine that Gemini can only explain, never invent.

USP: A defensible, auditable decision engine built on real clinical nutrition and drug-interaction logic, not a chatbot wrapper. Every alert traces back to the exact rule and data point that triggered it — the trust layer that makes this credible for a health-adjacent product, and hard for a weekend clone to replicate.

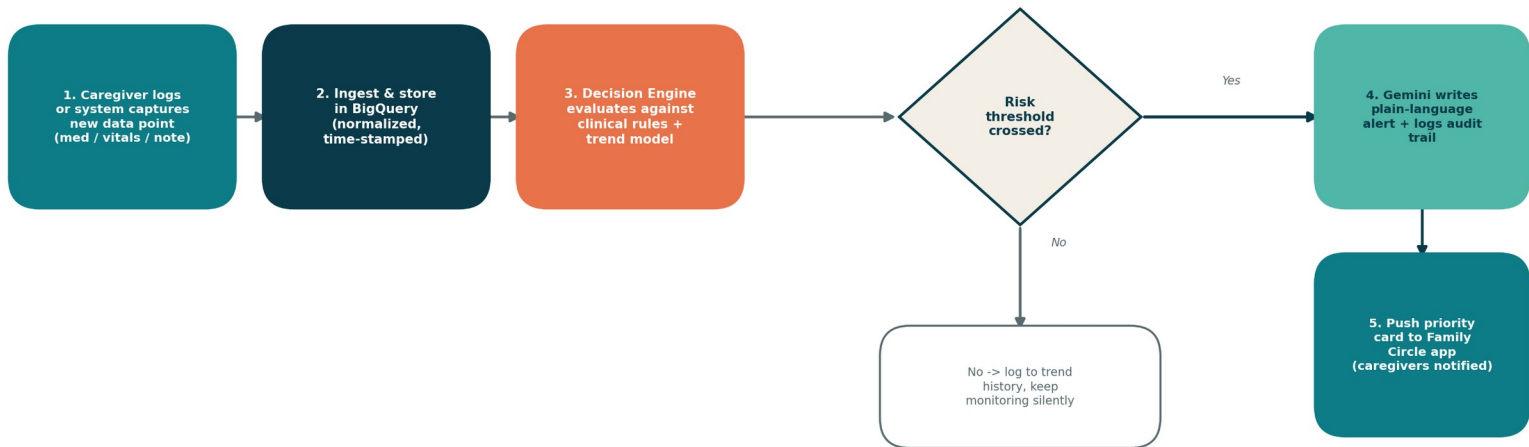
List of features offered by the solution

- **Multi-source data capture:** medication (photo -> OCR), home vitals, diet/meal notes, and caregiver observations, all in one place.
- **Real-time risk-scoring engine:** clinical rule logic plus trend/anomaly detection catches interactions and drift a caregiver would miss.
- **Plain-language priority alert cards:** Gemini turns a risk score into a short, non-clinician-friendly explanation of what changed and why it matters.
- **Full audit trail:** every alert traces back to the exact rule and source data point that triggered it -- nothing is a black box.
- **Family Circle sharing:** every caregiver in the family sees the same alert at the same time, so nothing falls through the cracks between siblings.
- **Historical trend view:** BigQuery time-series makes slow-building patterns visible weeks before they become an emergency.

Process flow diagram or Use-case diagram

Manual cross-check today: 15-30 min, often missed

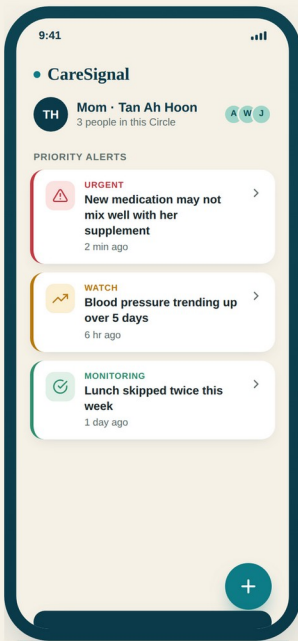
CareSignal end-to-end (steps 1→5): under 5 seconds



Wireframes/Mock diagrams of the proposed solution

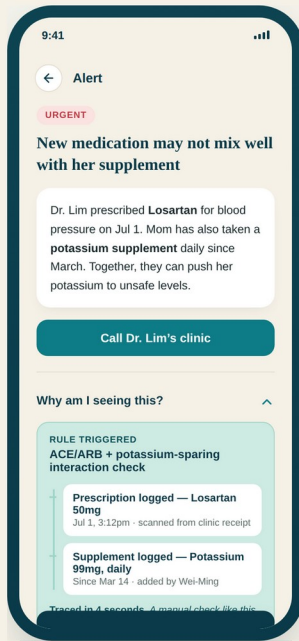
1. Family Circle home

Ranked alert cards, colour-coded by severity



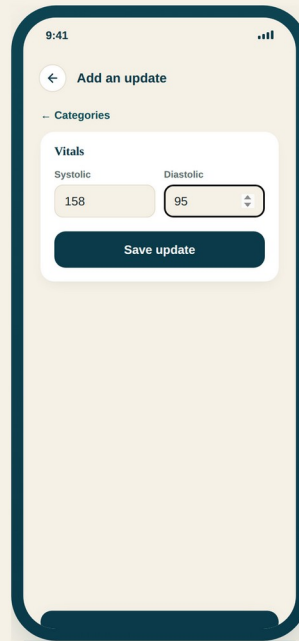
2. Alert detail

"Why am I seeing this?" → full audit trail

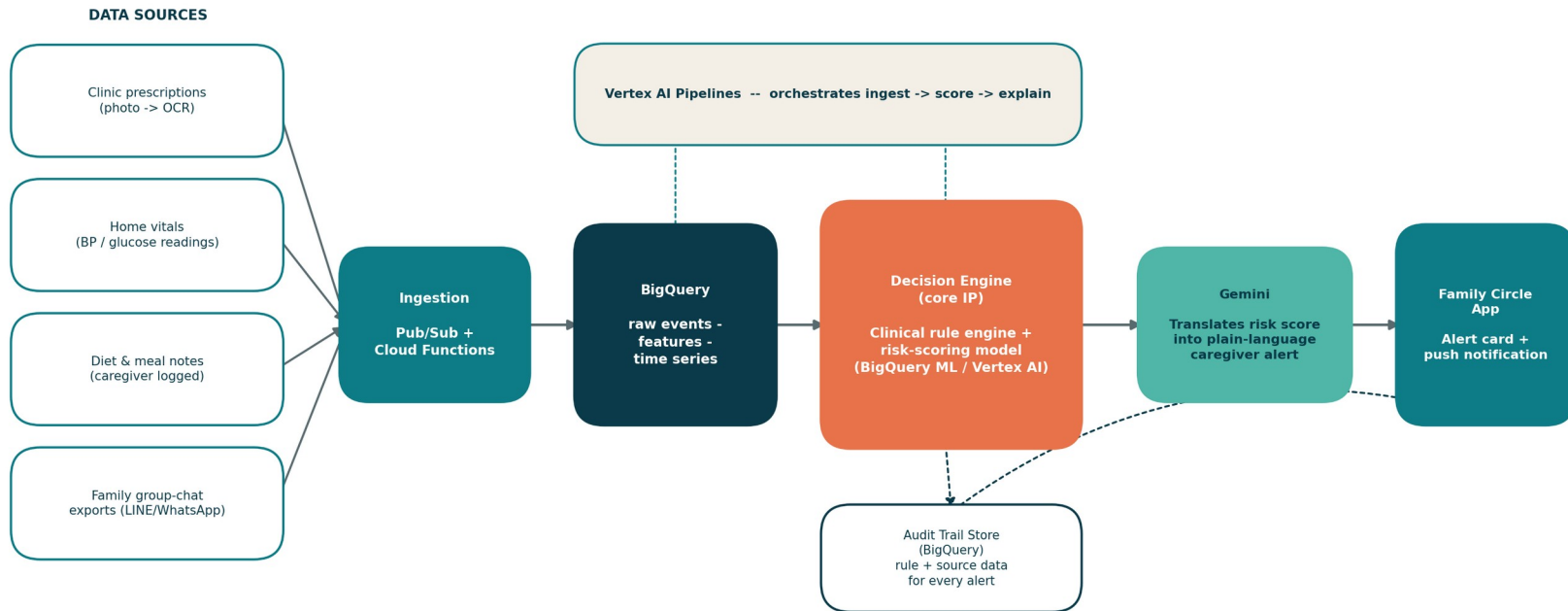


3. Quick log

Ten seconds to add a new reading



Architecture diagram of the proposed solution:



caregiver taps "why" -> drills down to the exact rule + data behind every alert

Technologies / Google / Nvidia Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

- **BigQuery:** central store for multi-source events, features and time-series trend detection; also runs the risk-scoring model via BigQuery ML for low-latency, serverless inference.
- **Cloud Functions + Pub/Sub:** the ingestion layer for heterogeneous inputs (OCR'd prescriptions, vitals entries, caregiver notes).
- **Vertex AI Pipelines:** orchestrates the ingest -> score -> explain flow so every run is repeatable and auditable.
- **Gemini:** the explanation layer only -- translates a risk score into a caregiver-readable alert; it never makes the clinical judgment itself.
- **Clinical rule validation:** 17 interaction rules, 14 sourced from a peer-reviewed expert-consensus panel (Phansalkar et al. 2012, JAMIA) plus a live openFDA label cross-check; all covered by 27 passing automated tests.

Why this stack: serverless components (Pub/Sub, Cloud Functions, BigQuery) scale from one family to an entire community without re-architecture, and keeping decision logic in BigQuery ML/rules rather than in a prompt keeps behavior consistent, explainable and auditable -- essential for anything health-adjacent.

Snapshots of the prototype

run_pipeline.py — real output

```
$ python3 run_pipeline.py

Scored subj_tan_ah_hoon against all rules in 6 ms.
Manual cross-check of the same data typically takes 15-30 min.

Found 3 alert(s):
[URGENT ] New medication may not mix well w/ potassium
[WATCH  ] Blood pressure trending up over 5 days
[MONITOR] Meals skipped 2 times this week
```

pytest — 27/27 passing

```
$ python3 -m pytest tests/ -v

test_decision_engine.py ..... 11 passed
test_onchigh_interactions.py ..... 6 passed
test_openfda_client.py ..... 9 passed

===== 27 passed in 0.4s =====
```



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Thank you

